

QUESTIONS 10 / 10 completed

Linear Search

LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total comparisons made, search status (Found/Not Found).

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 #include<time.h>
3 int main()
4 {
5     int a[]={45,23,67,12,89,34,56,78,90,11};
6     int n=10;
7     int key,i,found=0;
8     printf("enter target value:");
9     scanf("%d",&key);
10    clock_t start=clock();
11    for(i=0;i<n;i++)
12    {
13        if(a[i]==key){
14            found=i;
15            break;
16        }
17    }
18    clock_t end=clock();
19    double time=(double)(end-start)/CLOCKS_PER_SEC;
20    if(found)
21    {
```

INPUT

34

OUTPUT

```
enter target value:
target 34 found at positive 5
time:0.000000
```

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```
1 printf("enter target value: ");
2 scanf("%d",&key);
3 clock_t start=clock();
4 for(i=0;i<n;i++)
5 {
6     if(a[i]==key){
7         found=1;
8         break;
9     }
10 }
11 clock_t end=clock();
12 double time=(double)(end-start)/CLOCKS_PER_SEC;
13 if(found)
14 {
15     printf("\ntarget %d found at positive %d\n",key,i);
16 }
17 else
18     printf("target not found\n");
19 printf("time:%f",time);
20 }
```

OUTPUT

```
enter target value:
target 34 found at positive 5
time:0.000000
```

INPUT

34